

Code-Layout, Readability and Reusability

Date	2 June 2026
Team ID	XXXXXX
Project Name	EduGenie - AI-Powered Educational Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability

This document evaluates the quality of the **EduGenie** codebase in terms of its structure, readability, maintainability, and reusability. The project follows modular software engineering principles by separating the backend, frontend, AI services, and utility functions into independent components. Meaningful naming conventions, proper documentation, reusable functions, and consistent coding standards have been adopted throughout the project to improve code quality and simplify future enhancements.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform indentation and spacing maintained throughout the project.	Yes	PEP 8 coding standards followed.
2	Proper File Structure	Backend, frontend, templates, static files organized logically.	Yes	Modular structure. All project files are in their respective folders.

3	Meaningful Variable Names	Variables use descriptive purpose.	Yes names represent function	Easy to understand and maintain.
4	Function / Method Names	Functions clearly describe their operations.	Yes	Self-explanatory API methods
5	Code Comments	Important logic is documented using docstrings.	Yes	Critical modules documented.
6	Modular Design	Functionalities are divided into modules.	Yes	AI services separated into modules.
7	No Redundant Code	Duplicate code is minimized through reusable functions.	Yes	Helper functions reused across APIs.
8	Error Handling	API exceptions and runtime errors are handled gracefully.	Yes	Try-except blocks and HTTP status codes implemented.

Reusable Components/Modules

S.No	Component/ Module Name	Language/ Technology	Where Reused	Reusability Level
1	Question Answering Module	Python (Gemini API)	Educational Question Answering	High
2	Concept Explanation Module	Python (LaMini-Flan T5)	Simplified Topic Explanation	High
3	Quiz Generation Module	Python (Gemini API)	Automatic MCQ Generation	High
4	Text Summarization Module	Python (Transformers)	Educational Content Summarization	High
5	Learning Path Recommendation Module	Python (Gemini API)	Personalized Study Roadmaps	High
6	FastAPI Backend	FastAPI	Handles API requests for all modules	High
7	HTML Templates	HTML + Jinja2	Shared user interface pages	Medium
8	CSS Stylesheets	CSS	Common styling across all pages	Medium
9	Utility Functions	Python	Input validation, API response formatting, and helper methods	High
10	Configuration Module	Python (.env)	API key and environment configuration	High

Overall Code Quality Assessment

Aspect	Rating(1-5)	Comments
Code Layout & Structure	5	Well-organized modular project architecture.
Readability	5	Descriptive naming conventions and clean code.
Reusability	5	AI modules and helper functions are reusable across different components.

		multiple features.
Documentation/Comments	5	Important functions include comments and documentation.
Error Handling	5	Robust exception handling improves reliability.
OverallScore	5/5	Production-ready coding practices with scalable architecture.